<u>Recall</u> $\quad \mathrm{Ric}_{M^n} \geq -(n-1)\delta$

$$d(p, q_\pm) \geq L$$

$$E(p) = d(p, q_-) + d(p, q_+) - d(q_-, q_+) \leq \varepsilon$$

$$b_\pm(x) = d(x, q_\pm) - d(p, q_\pm)$$

$h_\pm$ def. by $\quad \Delta h_\pm = 0$ on $B_R(p)$

$\qquad\qquad\qquad\quad h_\pm = b_\pm$ on $\partial B_R(p)$

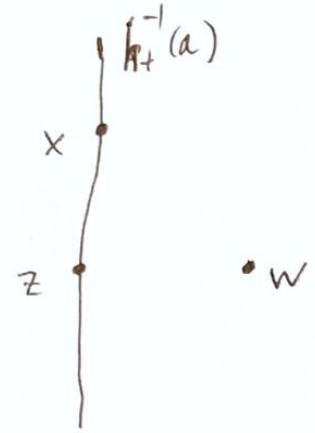
<u>Abresch – Gromoll</u> : $\quad E(x) \leq \Psi = \Psi(\delta, L^{-1}, \varepsilon \mid n, R)$ on $B_R(p)$

$$\Longrightarrow \quad |h_\pm - b_\pm| \leq \Psi$$

- $\displaystyle\fint_{B_R(p)} |\nabla h_+ - \nabla b_+|^2 \leq \Psi$, so $\displaystyle\fint_{B_R(p)} \big||\nabla h_+| - 1\big|^2 \leq \Psi$

- $\displaystyle\fint_{B_{R/2}(p)} |\mathrm{Hess}_{h_+}|^2 \leq \Psi$

<u>Lemma</u> Assume $(*)$. Let $x, z, w \in B_{R/8}(p)$ s.t. $x \in h_+^{-1}(a)$, $z \in h_+^{-1}(a)$ is a closest point to x on $h_+^{-1}(a)$



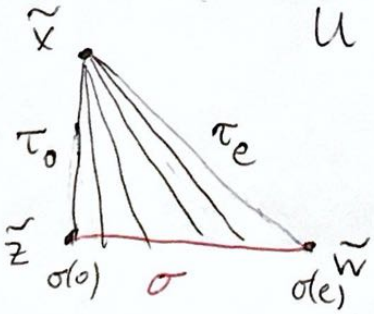
Then $\left| d(x,w)^2 - d(x,z)^2 - d(z,w)^2 \right| \leq \Psi$

<u>Proof</u> Apply iterated segment ineq. to $|\mathrm{Hess}_{h_+}|$

$\Longrightarrow$ can find $\tilde{x}, \tilde{z}, \tilde{w}$ s.t. $d(x,\tilde{x}), d(\tilde{z},z), d(w,\tilde{w}) < \Psi$

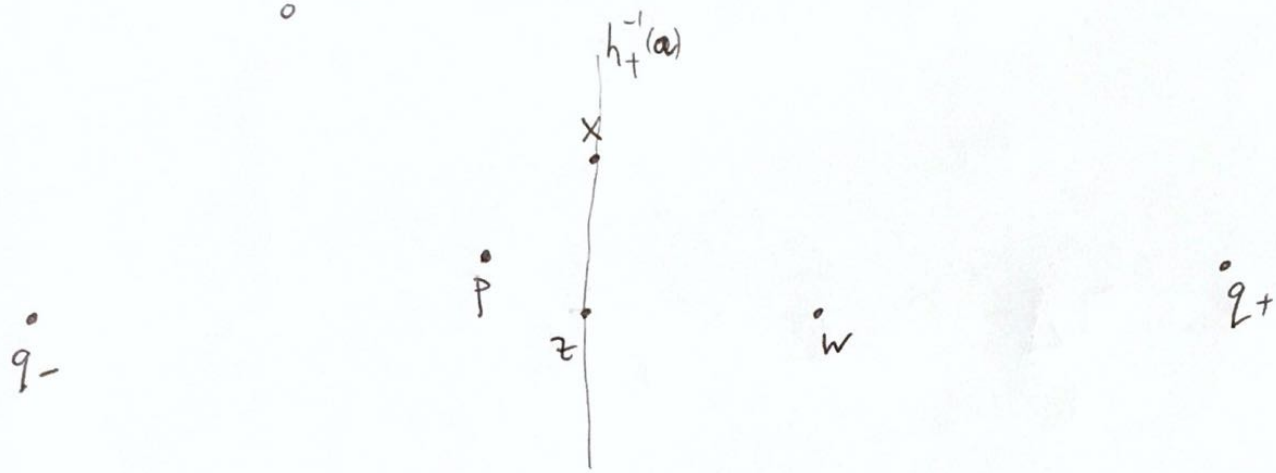
and if $\sigma : [0, e] \to M^n$ min. geodesic from $\tilde{z}$ to $\tilde{w}$ then for set $U \subset [0,e]$ of full measure for all $s \in U$ the min. geodesic $\tau_s : [0, \ell(s)] \to M^n$ joining $\tilde{x}$ to $\sigma(s)$ is unique

and
$$\int_U \int_0^{\ell(s)} |\mathrm{Hess}_{h_+}(\tau_s(t))| \, dt \, ds \leq \Psi$$



In addition by segment ineq. applied to $\big||\nabla h_+| - 1\big|$ we can assume

$$\int_0^e \big||\nabla h_+(\sigma(s))| - 1\big| \, ds \leq \Psi$$



Abresch-Gromoll $\Rightarrow$ $a = b_+(z) \leq b_+(w) + d(z,w)$

$$-b_+(w) + \Psi \geq b_-(w) \geq b_-(z) + d(z,w)$$
$$\geq -a + d(z,w) - \Psi$$

$$\Longrightarrow \quad \big| d(w,z) - |b_+(w) - b_+(z)| \big| \leq \Psi$$

$$\Rightarrow \quad \big| d(w,z) - |h_+(\tilde{w}) - h_+(\tilde{z})| \big| \leq \Psi$$

We assume $h_+(\tilde{w}) < h_+(\tilde{z})$

So $\big| d(w,z) - (h_+(\tilde{z}) - h_+(\tilde{w})) \big| \leq \Psi$

$$h_+(\tilde{w}) - h_+(\tilde{z}) = \int_0^e \nabla h_+(\sigma(s)) \cdot \sigma'(s) \, ds$$

$$\Rightarrow \quad \left| \int_0^e \nabla h_+(\sigma(s)) \cdot \sigma'(s) + 1 \, ds \right| \leq \Psi$$

so $$\int_0^e |\nabla h_+(\sigma(s)) + \sigma'(s)|^2 \, ds$$

$$= \int_0^e |\nabla h_+(\sigma(s))|^2 - 1 + 2\left(\nabla h_+(\sigma(s)) \cdot \sigma'(s) + 1\right) ds$$

$$\leq \Psi$$

For $t \in [0, \ell(s)]$

$$\left| \langle \nabla h_+(\tau_s(t)), \tau_s'(t) \rangle - \langle \nabla h_+(\tau_s(\ell(s))), \tau_s'(\ell(s)) \rangle \right|$$

$$= \left| \int_t^{\ell(s)} \mathrm{Hess}_{h_+}(\tau_s'(u), \tau_s'(u)) \, du \right| \leq \int_0^{\ell(s)} |\mathrm{Hess}_{h_+}(\tau_s(u))| \, du$$

$$\frac{1}{2} d(z,w)^2 = \frac{1}{2} d(\tilde{z}, \tilde{w})^2 \pm \Psi = \int_0^e s \, ds \pm \Psi$$

$$= \int_0^e h_+(\sigma(0)) - h_+(\sigma(s)) \, ds \pm \Psi$$

$$= \int_0^e h_+(\tau_s(0)) - h_+(\tau_s(\ell(s))) \, ds \pm \Psi$$

$$= \int_U \int_0^{\ell(s)} - \langle \nabla h_+(\tau_s(t)), \tau_s'(t) \rangle \, dt \, ds \pm \Psi$$

$$= \int_U \int_0^{\ell(s)} - \langle \nabla h_+(\tau_s(\ell(s))), \tau_s'(\ell(s)) \rangle \, dt \, ds \pm \Psi$$

$$= \int_U - \ell(s) \langle \underbrace{\nabla h_+(\underbrace{\tau_s(\ell(s))}_{\sigma(s)})}_{\sim -\sigma'(s)}, \tau_s'(\ell(s)) \rangle \, ds \pm \Psi$$

$$\left[\begin{array}{l}\text{variation of arclength : for all } s \in \mathcal{U} \\ \quad \ell'(s) = \langle \sigma'(s), \tau_s'(\ell(s)) \rangle\end{array}\right]$$

$$= \int_{\mathcal{U}} \ell(s)\ell'(s)\, ds \pm \Psi$$

$$= \frac{1}{2}\ell(e)^2 - \frac{1}{2}\ell(0)^2 \pm \Psi$$

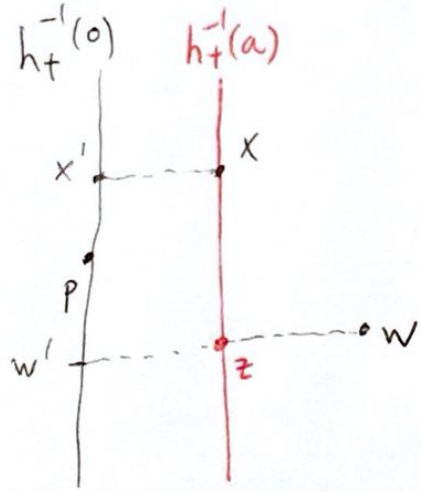
$$= \frac{1}{2}d(x,w)^2 - \frac{1}{2}d(x,z)^2 \pm \Psi$$

ie. $$d(z,w)^2 = d(x,w)^2 - d(x,z)^2 \pm \Psi \qquad \square$$

<u>Corollary</u> Let $h_+(p) = 0$, $X = h_+^{-1}(0) \subset M^n$

Then $$d_{GH}\left(B_{R/8}(p), B_{R/8}((0,p))\right) \leq \Psi$$
$$(0,p) \in \mathbb{R} \times X$$

<u>Proof</u> Let $F : B_{R/8}(p) \longrightarrow \mathbb{R} \times X$



$$w \longmapsto (h_+(w), z)$$

z = closest pt. to w on $h_+^{-1}(0)$

<u>Goal</u>: $d_{\mathbb{R}\times X}(F(x), F(w))$
$= d(x,w) \pm \Psi$